CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 13: **Our Environment** | Quick revision, one page

Ecosystem

- Ecosystem = biotic (producers, consumers, decomposers) + abiotic (light, temperature, water, soil) components. Natural (forest, pond) or artificial (garden, crop field, aquarium).
- Producers: green plants and some bacteria (autotrophs). Consumers: herbivores, carnivores, omnivores, parasites. Decomposers: bacteria and fungi (natural cleaners).

Food chain, web and energy flow

- Food chain: grass $\rightarrow$ grasshopper $\rightarrow$ frog $\rightarrow$ snake $\rightarrow$ hawk (arrows point towards the eater). Trophic levels: producer (T1), primary consumer (T2), secondary (T3), tertiary (T4).
- Food web: many interlinked food chains. Energy flow is unidirectional (Sun $\rightarrow$ producers $\rightarrow$ consumers).
- 10 per cent law: only about 10% of energy passes to the next trophic level; so energy at level n is $E_n = E_1 \times (0.1)^{n-1}$ and food chains have 3 to 4 trophic levels.
- Example: producer 10,000 J $\rightarrow$ primary 1000 J $\rightarrow$ secondary 100 J $\rightarrow$ tertiary 10 J.

Human activities and the ozone layer

- Ozone formation: $\mathrm{O_2} \xrightarrow{\mathrm{UV}} \mathrm{O} + \mathrm{O}$; $\mathrm{O} + \mathrm{O_2} \longrightarrow \mathrm{O_3}$. Ozone shields the Earth from UV radiation.
- CFCs (refrigerators, aerosols, fire extinguishers) deplete ozone; UNEP agreement in 1987 froze CFC production. Effects: skin cancer, cataract.

Waste

- Biodegradable (broken down by microorganisms): vegetable peels, paper, cotton, dung. Non-biodegradable: plastics, glass, metals, DDT.
- Biomagnification: non-biodegradable chemicals (pesticides) accumulate and increase along the food chain; highest in top consumers.
- Waste management: segregation (green and blue bins), reduce-reuse-recycle, composting, landfills.

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